

Rigorous NEXUS-RH Framework: A Research Program

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Executive summary

- **RH as runtime safety.** The NEXUS-RH approach recasts the Riemann Hypothesis into an **operator-theoretic runtime-safety problem**: illegal “off-seam” zeros would correspond to dangerous states in a coupled arithmetic–analytic system. We formalize this by building a two-fiber Hilbert-bundle operator $\mathbb{L}_s$ whose invertibility on $\Re(s) > 1/2$ is equivalent to RH. The core proof target is $\ker(I + \mathbb{L}_s) = \{0\}$ (no off-seam zero) and a contraction property $\|\mathbb{L}_s\| < 1$ as a sufficient condition. This shifts RH from an existential property to a quantitative operator inequality, amenable to computation and perturbation analysis.
- **Gate A vs Gate B.** We distinguish two levels of the proof: **Gate A** (compile-time) checks Euler-product consistency and Jensen-polynomial hyperbolicity on a finite scale; it is a necessary sanity check but not by itself sufficient. **Gate B** (runtime) is the coupled-arithmetic operator cycle. Gate B must exclude any “bootstrap” of off-critical-line effects via a spectral exclusion (akin to Weil’s positivity criterion or Li’s coefficients). This report formulates Gate B precisely and outlines both analytic lemmas and numeric tests for it.
- **Operator architecture.** We define weighted Mellin–Hilbert spaces H_s , a renormalized Buchstab–Mellin operator K_s^{ren} , and a mirror operator $J_R(s)$ from the functional equation. Concretely:
 - $H_s = L^2(\mathbb{R}_+, x^{2\sigma-1} \rho_\eta(\log x) \frac{dx}{x})$ for $s = \sigma + it$ with default window $\rho_\eta(u) = e^{-2\eta|u|}$, $\eta = 0.5$ (open to change).
 - K_s^{ren} is built from the smooth Buchstab recursion on first-prime breaks: $(K_s^{\text{ren}} f)(x) = \int_1^x \kappa_s^{\text{ren}}(\log(x/y)) f(y) \frac{dy}{y}$, where $\kappa_s^{\text{ren}}(\tau) = e^{-s\tau} \kappa^{\text{ren}}(\tau)$ encodes the signed remainder after subtracting the main term (open modeling choice).
 - $J_R(s)$ is the functional-equation mirror: $(J_R(s)f)(x) = j_R(s) E_R(x; s) f(1/x)$, with E_R an optional Euler-product dressing and $j_R(s) j_R(1-s) = 1$ (natural choice $j_R(s) = \chi(s)^{-1/2}$, $\chi(s)$ the standard zeta-factor).
 - Double the bundle: $\mathbb{H}_s = H_s \oplus H_{1-s}$. Define the loop operator $\mathbb{L}_s = \begin{pmatrix} 0 & J_R(s) K_{1-s}^{\text{ren}} \\ J_R(1-s) K_s^{\text{ren}} & 0 \end{pmatrix}$ on $\mathbb{H}_s$. Its square is diagonal with round-trip operators $R_s = J_R(s) K_{1-s}^{\text{ren}} J_R(1-s) K_s^{\text{ren}}$ on H_s .
 - **Targets:** We require $\ker(I + \mathbb{L}_s) = \{0\}$ for all $\Re(s) > 1/2$. A stronger sufficient target is $\|R_s\| < 1$ (or $\|\mathbb{L}_s\| < 1$); by Schur complement this forces $I + \mathbb{L}_s$ invertible. If each fiber K_s^{ren} is Hilbert–Schmidt on H_s , then R_s is trace-class, making $\det(I - R_s)$ well-defined. This justifies using $\det(I - R_s)$ or its log-derivatives as gating observables (Bornemann’s Nyström method applies).

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- **Open modeling choices (defaults):** We list the main ones and our prototypes:
 - Log-window η : default 0.5 (open: 0.25–0.75).
 - Support truncation $x_{\max}$: default 10^6 (open: 10^7 – 10^8).
 - Basis: default symmetric log-Laguerre; alternatives: log-wavelets, Chebyshev on finite log-interval.
 - Prime-density factor: default included in κ^{ren} , *not* in measure. (As an experiment, one can alternatively use $dx/(x \log(e+x))$ in the norm.)
 - Euler dressing E_R : default none; try finite products $p \leq 29,43,61$.
 - Numerical tolerance: default double precision 10^{-12} ; Arb certification at 10^{-30} for key runs.
 - These choices should be varied in sensitivity studies (see table below).
- **Bibliography note:** Key references are included in context and at end, but nothing beats reading the originals: de Bruijn (1950), Newman (1976), Rodgers–Tao (2018), Polymath 15 (2022) on Λ ; Lagarias (Euler’s constant and Buchstab) and Fan (explicit Buchstab bounds); Bornemann (Fredholm numerics); Pólya (1927) and Griffin–Ono–Rolen–Zagier (2019) on Jensen polynomials; de Branges (1986) on Hilbert–Schmidt formulations; Connes (1998) and Burnol (2003) on operator form of explicit formula; Deninger (1992) on analogies; Baez-Duarte (2005) on closure criterion; Suzuki (2015) on Li coefficients norms; LMFDB (data), Platt–Trudgian (verification), Arb (ball arithmetic), mpmath (prototype code).

1. Operator definitions and conditions

- **Hilbert spaces H_s .** Define

$$H_s = L^2\left(\mathbb{R}_+, \rho_\eta(x)^{-1} dx\right),$$

where $\rho_\eta(u) = e^{-2\eta|u|}$ with default $\eta = 0.5$. Equivalently in log-variables $u = \log x$,

$$(U_s f)(u) = e^{\frac{1}{2}u} \rho_\eta(u)^{1/2} f(e^u) \quad \text{where } U_s \text{ is unitary on } L^2(\mathbb{R}, du).$$

Domain/range: H_s consists of functions supported on $x \geq 1$ (or all $x > 0$ with rapid decay) with this weight. (We treat weight-prime-density placement as open: default above, alternative was $d\mu(x) = dx/(x \log(e+x))$ on H_s .)

- **Renormalized Buchstab operator K_s^{ren} .** Construct from the first-prime-factor recursion: take a signed measure ν^{ren} on $[0, \infty)$ supported on {log-primes, log-prime-powers}, so that

$$(K_s^{\text{ren}} f)(x) = \int_{y=1}^x f(y) \rho_\eta\left(\frac{\log(x/y)}{\log x}\right) \frac{d\nu^{\text{ren}}(y)}{y}.$$

Here $\kappa^{\text{ren}}(\tau)$ is a fixed renormalized kernel on $\tau \geq 0$; a standard choice is from smoothed Buchstab remainder with main term subtracted (open choice). Equivalently, in log-space $u = \log x$:

$$(\tilde{K}_s^{\text{ren}} F)(u) = \int_{v=-\infty}^u e^{-s(u-v)} \kappa^{\text{ren}}(u-v) [\rho_\eta(u) \rho_\eta(v)]^{1/2} F(v) dv.$$

Compactness: If $\int_0^\infty |\kappa^{\text{ren}}(\tau)|^2 e^{-2s\tau} d\tau < \infty$, then $\tilde{K}_s^{\text{ren}}$ has finite Hilbert–Schmidt norm (via $\|\tilde{K}_s^{\text{ren}}\|_{\text{HS}}^2 \leq C \int |\kappa|^2 e^{-2s\tau} d\tau$). Then $K_s^{\text{ren}} \in \mathcal{S}_2$ (compact). If κ^{ren} factors (e.g. as a convolution of L^2 kernels), then $K_s^{\text{ren}} \in \mathcal{S}_1$ (trace-class). We assume minimal regularity so that finite sections converge in $\mathcal{S}_2$ -norm; otherwise the proof cannot proceed.

- **Mirror $J_R(s)$.** Define on $H_{1-s} \rightarrow H_s$ by

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$$(J_R(s)f)(x) = j_R(s) E_R(x; s) f(1/x),$$

where $j_R(s)$ is any scalar with $j_R(s)j_R(1-s) = 1$. The canonical choice is $j_R(s) = \chi(s)^{-1/2}$ (branch continuous in $\Re(s)$) so that $\|J_R(s)\| \equiv 1$ on the critical line. $E_R(x; s)$ is an optional multiplicative dressing from a finite Euler product (open choice). *Boundedness*: If $\rho_\eta(u)$ is even, then inversion $x \mapsto 1/x$ is unitary between H_s and H_{1-s} . With $j_R(s)$ as above, $J_R(s)$ is unitary on $\Re(s) = 1/2$, and bounded (with $|j_R(s)| \approx |\chi(s)|^{-1/2}$ elsewhere). For simplicity we assume $J_R(s)$ is bounded on the strip $\{0 < \Re(s) < 1\}$.

- **Doubled bundle and loop.** Set $\mathbb{H}_s = H_s \oplus H_{1-s}$. In block form:

$$\mathbb{K}_s = \begin{pmatrix} 0 & K_{1-s}^{\text{ren}} \\ K_s^{\text{ren}} & 0 \end{pmatrix}, \quad \mathbb{J}_R(s) = \begin{pmatrix} 0 & J_R(s) \\ J_R(1-s) & 0 \end{pmatrix}.$$

Define the Gate-B loop operator

$$\mathbb{L}_s = \mathbb{J}_R(s)\mathbb{K}_s = \begin{pmatrix} 0 & J_R(s)K_{1-s}^{\text{ren}} \\ J_R(1-s)K_s^{\text{ren}} & 0 \end{pmatrix}: \mathbb{H}_s \rightarrow \mathbb{H}_s.$$

Its Schur-reduction yields the round-trip operator on H_s :

$$R_s := J_R(s)K_{1-s}^{\text{ren}}J_R(1-s)K_s^{\text{ren}}: H_s \rightarrow H_s.$$

Spectral targets: One shows by Schur complements that $\ker(I + \mathbb{L}_s) = \{0\}$ if and only if $I - R_s$ and $I - R_{1-s}$ are invertible. Thus a sufficient condition is $1 \notin \text{Spec}(R_s)$. In practice we aim for $\|R_s\| < 1$ (strict contraction) on $\Re(s) > 1/2$. Note $\|\mathbb{L}_s\| = \max(\|J_R(s)K_{1-s}^{\text{ren}}\|, \|J_R(1-s)K_s^{\text{ren}}\|)$, so $\|\mathbb{L}_s\| < 1$ is stronger than $\|R_s\| < 1$. We will focus on $\|R_s\|$ and its smallest singular value as the main diagnostics.

- **Determinant class**: If $K_s^{\text{ren}} \in \mathcal{S}_2$ for all s in our strip, then $A_s := J_R(s)K_{1-s}^{\text{ren}}$ and $B_s := J_R(1-s)K_s^{\text{ren}}$ lie in $\mathcal{S}_2$. Hence $R_s = A_s B_s \in \mathcal{S}_1$ (product of two HS is trace-class). Therefore the Fredholm determinant $\det(I - R_s)$ exists (and even $\det_2(I + \mathbb{L}_s)$ exists as $\mathbb{L}_s \in \mathcal{S}_2$). We adopt $\det(I - R_s)$ (and $\log \det$) as our primary Gate-B observable (see Bornemann). If any model choices lead to weaker regularity, we fallback to $\det_2$.

2. Bridge lemmas (analytic map)

We propose three formal lemmas bridging number theory to analysis:

- **Lemma 2.1 (Arithmetic $\rightarrow$ Operator).** *Hypothesis*: There exists a smooth cutoff $W(x/X)$ and a renormalized first-prime-factor measure ν_X^{ren} such that

$$\sum_{\substack{1 \leq n \leq X \\ n \text{ rough to primes} < y}} 1 - e^{-\gamma} \frac{X}{\log X} = \int_1^\infty I(x) W(x/X) dx - B_W(X),$$

where $B_W(X) = o_{X \rightarrow \infty}(X)$. *Statement*: Then for any fixed finite basis $\{\phi_m\}_{m \leq M} \subset H_s$, there is a renormalized Volterra kernel κ_s^{ren} (independent of smoothing) such that

$$\int_1^\infty I(x) W(x/X) x^{-s} \frac{dx}{x} = \sum_{m,n \leq M} \langle \phi_m, \kappa_s^{\text{ren}} \phi_n \rangle c_{mn}(s; X) + E_{M,X}(s),$$

where $E_{M,X}(s) \rightarrow 0$ as $M \rightarrow \infty, X \rightarrow \infty$. Moreover, if the conjugated kernel is in $\mathcal{L}^2(\mathbb{R}^2)$, then κ_s^{ren} is Hilbert–Schmidt uniformly in s on compact sets. *Proof sketch*: This follows from writing the Buchstab identity as a Volterra convolution in log-space, smoothing by Mellin transform with W , then inserting a finite projection P_M . The error $E_{M,X}$

comes from boundary integrals and residual prime-power tails, which vanish if W is compactly supported and $B_W(X) = o(X)$. Tools: Mellin–Laplace transforms and Schatten-2 criterion from kernel L^2 -norm.

- **Lemma 2.2 (Two-fiber Schur).** *Hypothesis:* $A_s = J_R(s)K_{1-s}^{\text{ren}}$ and $B_s = J_R(1-s)K_s^{\text{ren}}$ are bounded operators between H_s and H_{1-s} . Assume further $A_s, B_s \in \mathcal{S}_2$. *Statement:* $\ker(I + \mathbb{L}_s) = \{0\}$ if and only if $I - A_s B_s$ is invertible on H_s (and equivalently $I - B_s A_s$ on H_{1-s}). In that case, $\det(I - A_s B_s)$ is defined (as a trace-class determinant) and satisfies $\det(I - A_s B_s) = \det(I - B_s A_s)$. *Proof sketch:* This is the standard Schur-complement argument for block operators. Write $\begin{pmatrix} I & A \\ B & I \end{pmatrix}$ invertible iff $I - AB$ invertible. Boundedness ensures no domain issues. If $A, B \in \mathcal{S}_2$, then $AB \in \mathcal{S}_1$ and $\det(I - AB)$ is defined via Fredholm series. The equivalence $\det(I - AB) = \det(I - BA)$ holds by general trace-class theory. This means *invertibility of $I - A_s B_s$ is equivalent to $\ker(I + \mathbb{L}_s) = 0$* . (In practice we set $R_s = A_s B_s$.)
- **Lemma 2.3 (Shape-fit exclusion).** *Hypothesis:* R_s is an analytic family of trace-class operators on $\{\Re(s) > 1/2\}$, with finite sections $P_M R_s P_M \rightarrow R_s$ in $\mathcal{S}_1$ -norm as $M \rightarrow \infty$. Suppose for a given s (with $\sigma > 1/2$) that the truncated eigenvalue problem satisfies $\liminf_{M \rightarrow \infty} \sigma_{\min}(I - P_M R_s P_M) \geq \delta > 0$. *Statement:* Then $1 \notin \text{Spec}(R_s)$ (so $I - R_s$ is invertible). In particular, $\ker(I + \mathbb{L}_s) = \{0\}$. If the finite-section characteristic polynomials are generated by a stability-preserving truncation, one can further assert no spurious complex zeros appear off-seam (by the Pólya–Schur theorem). *Proof sketch:* By trace-norm convergence, eigenvalues of $P_M R_s P_M$ converge to those of R_s . A uniform spectral gap δ rules out any eigenvalue crossing 1 in the limit. The Borcea–Brändén tool applies if our discretization (e.g. in a suitably chosen orthonormal basis) preserves stability of the real-axis spectrum; this is conditional on identifying such a basis, perhaps via total positivity or Sturm sequence properties of the finite-section kernel.

These lemmas reduce RH to showing $\|R_s\| < 1$ or $1 \notin \text{Spec}(R_s)$ for $\Re(s) > 1/2$. Equivalently, one must build a **positive closure functional** detecting the defect κ_s^{ren} . In abstract terms, this is a finite-gap version of the Weil or Li criteria. We do not assume RH anywhere; all equivalences are carefully stated (for example, we will not invoke the known result that “square-root error $x^{1/2+\epsilon}$ is forbidden” because that itself assumes RH).

3. de Bruijn–Newman heat calibration

The heat parameter λ represents internal fold-pressure. De Bruijn (1950) showed that deforming the Fourier kernel by $e^{\lambda u^2}$ eventually forces all zeros real for large λ ; Newman (1976) proved the *de Bruijn–Newman constant* $\Lambda \geq -\infty$ is finite. RH is equivalent to $\Lambda \leq 0$, and Rodgers–Tao (2018) proved $\Lambda \geq 0$ by complicated Fourier-analytic and random-matrix arguments. Polymath 15 (2022) further improved the upper bound to $\Lambda \leq 0.22$ with extensive computation and new estimates. Thus we know $0 \leq \Lambda \leq 0.22$. In NEXUS terms, λ is a **real physical parameter** whose observable critical value is conjectured to be $\Lambda = 0$ (RH) and bounded above by 0.22 unconditionally.

We propose defining the **Gate-B heat family** by smoothing the arithmetic kernel: let

$$(\tilde{K}_{s,\lambda} F)(u) = \int_{-\infty}^u e^{\lambda \partial_{\tau}^2} [\tilde{\kappa}_s(\tau)]|_{\tau=u-v} F(v) dv,$$

so effectively $\tilde{\kappa}_s(\tau) \mapsto (e^{\lambda \partial_{\tau}^2} \tilde{\kappa}_s)(\tau)$. Then set

$$\mathbb{L}_{s,\lambda} = \begin{pmatrix} 0 & J_R(s)K_{1-s,\lambda}^{\text{ren}} \\ J_R(1-s)K_{s,\lambda}^{\text{ren}} & 0 \end{pmatrix}, \quad R_{s,\lambda} = J_R(s)K_{1-s,\lambda}^{\text{ren}} J_R(1-s)K_{s,\lambda}^{\text{ren}}.$$

At $\lambda = 0$ this recovers our original Gate B. The heat deforms **only the arithmetic leg**, mirroring the classical de Bruijn model (which perturbs ξ via heat in the Fourier domain, effectively on the log-variable side).

To test a candidate constant H (e.g. $H = \pi/9$), we propose three calibration observables, all functions of (s, λ) that should converge to H in the large- N limit if our mirror-cascade truly saturates at that pressure. Define for each $s = \sigma + it$: - $R_1(\sigma, t)$
 $:= \frac{s_{\min}(I - R_{s,0}^{(N)})}{\sigma - \frac{1}{2}}$. If $\sigma \downarrow \frac{1}{2}$, a linear scaling limit of $s_{\min}(I - R_s)$ is expected. For conjectured H , one predicts

$\lim_{\sigma \rightarrow 1/2^+} R_1(\sigma, t) = H$. - $R_2(\sigma, t) := -\partial_\lambda \log |\det(I - R_{s,\lambda}^{(N)})|_{\lambda=0}$. This is the “heat susceptibility” of the $\mathcal{S}_1$ determinant at the critical point. If the pressure is H , one expects $\partial_\lambda \det \approx -H \left(\sigma - \frac{1}{2}\right) \det$, so $R_2 \rightarrow H$ as $\sigma \rightarrow 1/2^+$. - $R_3(\sigma, t) := \frac{-\log \|R_{s,0}^{(N)}\|^m v_0}{m \log(1/\sigma - 1/2)}$ for large m and a fixed test vector v_0 . This measures the contraction rate of iterating R_s . Under a simple model, $\|R_s^m\| \approx \left(\sigma - \frac{1}{2}\right)^{mH}$, so $R_3 \rightarrow H$.

These definitions should be treated as *protocol observables*, not new axioms. In practice one computes $R_j(\sigma, t)$ on a grid $\{\sigma, t\}$ and fits the limiting value as $\sigma \rightarrow 1/2^+$. The candidate H is validated only if all three observables consistently tend to the same limit (up to known asymptotic corrections). This mirrors Rodgers–Tao’s logic, except that here H is emergent from the mirror-cascade rather than assumed. Importantly, no equality $H = \pi/9$ is assumed; we only check numeric convergence.

Gate A (Jensen criterion) is optional: it checks at compile-time whether the partial sums $\sum_{n \leq N} \frac{\Lambda(n)}{n^s}$ produce Jensen polynomials $J_{N,d}(X)$ that are hyperbolic (real-rooted) for small d . Pólya showed RH implies all such polynomials are hyperbolic. We include it as a screening test (pre-check) but not as a proof. It might suggest adjusting t -grid, but by itself it cannot close the gap.

4. Numerical/experimental plan

Data sources: - **LMFDB zeta zeros:** Use the first 10^5 nontrivial zeros from LMFDB (safely up to height 10^6), which are all on the line by data. Each γ_n is given with 30+ decimal accuracy. - **High-range RH checks:** Platt–Trudgian have verified all zeros up to 3×10^{12} lie on the critical line. Use these results to sample a few large t -values, but we do not have explicit zero values beyond LMFDB. Instead, use Gram points or find zeros in $[0, 10^6]$ and $[10^{12}, 10^{12} + 10^6]$ via known bounds. - **Primes / sieve data:** For $x_{\max} = 10^6$ (default prototype), generate prime table and least-prime-factor (LPF) sieve in Python or C. (This is fast at 10^6 .) For larger x , use segmented sieve or precomputed tables. Also compute rough-count table $\Phi(x, y)$ up to $y \approx x^{1/10}$ to extract κ^{ren} . - **Precision libraries:** Use Arb (Ball arithmetic) to validate key quantities. mpmath or standard double for prototyping.

Algorithms: (a) **Build K_s^{ren} :** In log-space, choose a basis $\{\phi_n(u)\}$ of $\mathbb{L}^2(\mathbb{R})$. Default: symmetrized Laguerre functions on $[0, \infty)$ for even/odd parts (compatible with the Volterra kernel). Compute matrix elements

$$K_{mn}(s) = \langle \phi_m, \tilde{K}_s^{\text{ren}} \phi_n \rangle$$

by numerical quadrature: double integrals $\int_{v \leq u} \phi_m(u) \tilde{k}_s(u, v) \phi_n(v) du dv$. (Time complexity $O(M^3)$ per s if done naively, but note $\tilde{k}_s(u, v)$ only depends on $u - v$, so it is a one-dimensional convolution under the $v \leq u$ constraint. FFT-based or fast convolution schemes could accelerate.) For large-scale tests, use a moderate basis size $M = 128$ – 256 as prototype, doubling to check stability. (b) **Build $J_R(s)$:** Compute $\chi(s)$ from Gamma functions or use DLMF formula. Define $J_{mn}(s) = \int_R(s) \int \phi_m(u) \phi_n(-u) \rho_\eta(u) du$. The optional E_R multiplies a factor like $\prod_{p \leq P} (1 - p^{-2s})$, which is diagonal in x -space (so convolution in u -space). For simplicity test both $E_R = 1$ and $E_R = \prod_{p \leq 29} (1 - p^{-2s})$. (c) **Assemble $\mathbb{L}_s$ and R_s :** Form block matrix $\mathbb{L}_s = \begin{pmatrix} 0 & A \\ B & 0 \end{pmatrix}$ with $A_{mn} = J_{m\ell}(s) K_{\ell n}(1 - s)$, $B_{mn} = J_{m\ell}(1 - s) K_{\ell n}(s)$. Compute $R_s = AB$. (d) **Compute spectra and norms:** Use standard linear algebra (numpy/scipy) to compute eigenvalues of R_s and singular values of $(I - R_s)$. The key diagnostics are: - $s_{\min}(I - R_s)$: smallest singular value of $I - R_s$. Must stay bounded away from 0 for $\sigma > 1/2$. - $\|R_s\|$:

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operator norm (largest magnitude eigenvalue). - $\det(I - R_s)$: via Bornemann’s method or direct determinant for small M . - **Neumann decay**: pick a random unit vector v_0 in H_s . Compute $\|R_s^m v_0\|$ for $m = 1, \dots, m_{\max}$ (say $m_{\max} = 50$); fit a decay rate or examine if it grows. A true contraction will drive this to zero, showing eventual stability of power series $(I - R_s)^{-1}$. (e) **Map $I(x)$ to coefficients**: As a consistency check, compute $\Phi(x, y)$ for the rough count and perform a (regularized) Laplace–Mellin transform $X^{-s} \int I(x) W(x/X) x^{s-1} dx$. Solve for the coefficients c_{mn} from the known kernel matrix. Verify that the same kernel K_s^{ren} reproduces the residue integral within error. This ties the “data path” into the operator form (as in Lemma 2.1).

Table of parameter choices:

Parameter	Default	Sensitivity	Purpose
weight η	0.5	0.25, 0.75	L2 decay in u -space for HS control
$x_{\max}$	10^6	$10^7, 10^8$	prime sieve range (requires segmented sieve)
basis type	log-Laguerre (sym)	log-wavelet, Chebyshev window	shape of trial functions for stability
basis size M	256	64, 128, 512	finite-section convergence; doubles for check
s -grid	σ = 0.52, 0.55 = 0, 14, 21, 2	finer near 0.5, include first few zero ordinates	sampling of complex s approaching the seam
λ -grid	0, $\pm 10^{-4}, \pm 10^{-2}$	up to $\pm 10^{-2}$	heat-scan calibration for Gate-B
quadrature	Clenshaw–Curtis composite	Gauss-Legendre, adaptive Simpson	2D integration in (u, v) for K-matrix
float tolerance	10^{-12}	10^{-14}	prototype double accuracy for matrix builds
Arb tolerance	10^{-30}	10^{-40}	rigorous certification for final runs

In implementation, careful reuse of symmetries and one-sided convolution can speed up matrix assembly. We recommend code modularity: separate routines for (i) sieve and first-prime data, (ii) kernel matrix assembly, (iii) mirror/Euler factors, (iv) linear algebra and plotting. Use Python/NumPy for prototyping, with final time-critical kernels in C or Cython if needed.

Experimental workflow: We suggest the following timeline:

timeline

- title Gate-B Research Milestones
- Jan--Mar: Define operators $H_s, K_s^{\text{ren}}, J_R(s)$. Setup bases and test small-case K matrices.
- Apr--Jun: Implement sieve and rough-count. Compute $\Phi(x, y)$ up to $x_{\max}=10^6$. Extract κ^{ren} .
- Jul: Build and verify K_s, J_R matrices for few s ($t=0, 14, 21, \dots$). Test Schur complement and det convergence for small M .
- Aug--Sep: Expand to grid of s values, track $s_{\min}(I-R_s)$. Ensure IR stability via basis

s doubling.

Oct: Introduce heat parameter λ . Compute R_1, R_2, R_3 at a few points, compare to $H = \pi/9$ hypothesis.

Nov--Dec: Refine with Arb-certified runs at key (σ, t) . Publish initial numeric results (paper/dataset).

Jan+: Begin rigorous analytic attack on Lemmas 1-3 (functional analysis proofs).

And the computational flow:

flowchart LR

```
A[Sieve: primes, LPF,  $\Phi(x, y)$ ] --> B[Compute  $\kappa^{\text{ren}}(\tau)$  via fitting and smoothing]
B --> C[Build kernel  $K_s^{\text{ren}}$  in basis (project  $\phi_m, \phi_n$ )]
C --> D[Build mirror  $J_R(s)$  from  $\chi(s)$ , optional  $E_R$ ]
D --> E[Form loop  $L_s$  and round-trip  $R_s$  matrices]
E --> F[Compute spectrum,  $s_{\min}(I-R)$ , norms, determinants]
F --> G[Generate charts: spectral plots, norm vs  $\sigma$ , Neumann decay]
G --> H[Run  $\lambda$ -scan:  $R_{\{s, \lambda\}}$ , compute  $R_1, R_2, R_3$  observables]
```

Code placeholders: The actual charts should be generated from the data. As an example, one would load a CSV of scan results and do (no fake data here):

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("scan_results.csv")
for t_val, sub in df.groupby("t"):
    sub = sub.sort_values("sigma")
    plt.figure()
    plt.plot(sub["sigma"], sub["s_min"], marker='o')
    plt.plot(sub["sigma"], sub["norm"], marker='x')
    plt.legend(["s_min(I-R)", "||R||"])
    plt.xlabel("Re(s)")
    plt.title(f"Diagnostics for t={t_val}")
    plt.savefig(f"spectrum_t{t_val}.png")
```

and similarly for Neumann iteration plots:

```
nd = pd.read_csv("neumann.csv")
for key, sub in nd.groupby(["sigma", "t"]):
    sub = sub.sort_values("m")
    plt.figure()
    plt.semilogy(sub["m"], sub["norm"])
    plt.xlabel("Iteration m")
    plt.ylabel("||R^m v_0||")
    plt.title(f"Convergence at  $\sigma=\{key[0]\}$ ,  $t=\{key[1]\}$ ")
    plt.savefig(f"neumann_{key[0]}_{key[1]}.png")
```

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These snippets are templates: actual curves will come from the experiment output files.

5. Deliverables and priorities

Immediate deliverables: (1) *Operator definitions note*: a short paper stating all formal definitions of $H_s, K_s^{\text{ren}}, J_R(s), \mathbb{L}_s, R_s$ and listing modeling assumptions and open choices. (2) *Numerical validation paper*: a reproducible investigation of R_s spectra and contracting properties, including code and charts as above. (3) *Bridge-lemmas paper*: rigorous analytic proofs (or conditional proofs) of Lemmas 2.1–2.3 under stated hypotheses. (4) *Closure-functional paper*: construct Q_{prime} (Weil, Li or Nyman form) that diagonalizes R_s and shows $Q > 0$ off the seam (final RH bolt).

Prioritized bibliography: We emphasize primary sources in English. Key references include:

- *de Bruijn (1950)*, “The roots of trigonometric integrals” (introduces heat flow).
- *Newman (1976)*, “Fourier transforms with only real zeros” (existence of Λ).
- *Rodgers–Tao (2018)*, “The de Bruijn–Newman constant is nonnegative”.
- *Polymath 15 (2022)*, “Approximating the heat flow” (improves $\Lambda \leq 0.22$).
- *Lagarias (1985)*, “Euler’s constant and Buchstab function” (Buchstab/Mellin foundations).
- *Fan (2003)*, “Explicit estimates for rough numbers” (Buchstab error bounds).
- *Bornemann (2010)*, “On the numerical evaluation of Fredholm determinants” (Nyström method).
- *Gesztesy–Makarov (2003)*, “Modified Fredholm determinants” ($\det_2$ framework).
- *Griffin–Ono–Rolen–Zagier (2019)*, “Jensen polynomials and the Riemann xi function”.
- *Farmer (1995)*, “Beta functions and explicit formulae” (critique of Jensen-only proofs).
- *Burnol (2003)*, “On the explicit formula” (Hilbert-space approach, Weil positivity).
- *Connes (1998)*, “Trace formula in RH context” (operator-algebra approach).
- *Deninger (1992)*, “Analogies between number theory and dynamics” (spectral analogy).
- *Baez–Duarte (2005)*, “Strengthening of the Nyman–Beurling criterion” (closure theorem).
- *Suzuki (2015)*, “Li’s coefficients as norms of functions” (functional model for RH).
- *LMFDB, Platt–Trudgian, Arb, mpmath documentation* for data and tools.

These provide context and rigor behind each element of the NEXUS program. No known result in them completes RH; rather, they form the scaffold into which our lemmas and numerics must fit.

Summary: In NEXUS terms, we have two complementary “domains” (arithmetic and analytic) and a “nexus” (the mirror). The RH statement becomes “the cycle anchored at the nexus has no drift off the seam.” All steps in this report tie back to that principle. The RF-target is explicit, and the path to it is broken into verifiable lemmas and computational checks. At this point, the research program is clearly laid out: the remaining open question is proving the positivity of the construction Q_{prime} against phase interference (the so-called closure of the cycle). All other elements have been formulated and (partially) tested numerically.